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THE HONG KONG POLYTECHNIC UNIVERSITY

DEPARTMENT OF COMPUTING

EXAMINATION

Course : Broad Discipline of COMP-61431

Subject : COMP3438 System Programming

Group : 1011

Session : 2018 / 2019 Semester I

Date : 20 December 2018 Time : 12:30 - 14:30

Time Allowed: 2 Hours Subject Lecturer: Dr LI Shuai

This question paper has 2 pages.

Instructions to Candidates:

1. Close books and notes.
2. Answer ALL questions.

Do not turn this page until you are told to do so!

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Question 1 [20%]

The inode structure of a simple file system has following properties:

e) The total size of an inode is 64 bytes.
f) Out of 64 bytes, 48 bytes are used to store meta data (filename, size, creation time etc.).
g) The remaining 16 bytes contain equal number of direct pointers and single indirect pointers. There are no double or triple indirect pointers.
h) Each pointer is four bytes in size.

Assume that each data block is 4KB in size.

Answer the following questions:

1. How many direct and single indirect pointers are present in the inode? (4%)
2. What is the maximum file size that we can have on this simple file system? (4%)
3. Suppose there is a file with maximum possible size on this file system. You are asked to read "*count*" number of bytes starting from byte location "*offset*" from that file. Where *offset = 0* mean (1st byte) start of the file, and offset=10 (say) means that starting from (10+1=)11th byte. Write down the list of pointers you will need to access the required data.

   For example: if offset = 50 and count = 100, this means you want to read data from byte 51 to byte 150. To read this data you will only need to access 1st direct pointer. By keeping in view of the first two examples below, fill the table (12%)

	Offset	Count	Pointers
Example 1	4000	1000	1st Direct pointer 2nd Direct pointer
Example 2	8000	2000	2nd Direct pointer *1st single indirect pointer -> direct pointer[1]
(1)	10000	10000	
(2)	400000	5000	
(3)	3000000	20000	

* "-> direct pointer[1]" represents first direct pointer inside 1st single indirect pointer.

Question 2 [10%]

Write down a sketch of the code to implement bi-directional communication between a parent and a child process. (No need to include header files).

Question 3 [10%]

Write down the major differences between polling vs interrupt-based device drivers. Which approach is better in your opinion?

Question 4 [20%]

Let $\sum = \{a, b, c, d\}$. Write down the regular expression of a string which contains

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1. Exactly one occurrence of "aabb". (10%)
2. One or more occurrences of "aabb". (10%)

Question 5 [20%]

Let $\sum = \{a, d\}$. Given the regular expression (a|b)+

1. Draw the NFA of this regular expression. (10%)
2. Convert the NFA to DFA and draw the transition graph. (10%)

Question 6 [20%]

Given the following grammar,

N = {S, M, N},
T = {a, b},
S = S,
P:

$S \rightarrow a\ S\ M\ N\ b \mid a \mid \varepsilon$
$M \rightarrow S \mid b$
$N \rightarrow M \mid a$

Answer the following questions

1. Write a leftmost derivation of this grammar. (6%)
2. Draw the parse tree for this grammar. (6%)
3. Find the FIRST set for all the symbols. (8%)

----------**END**----------